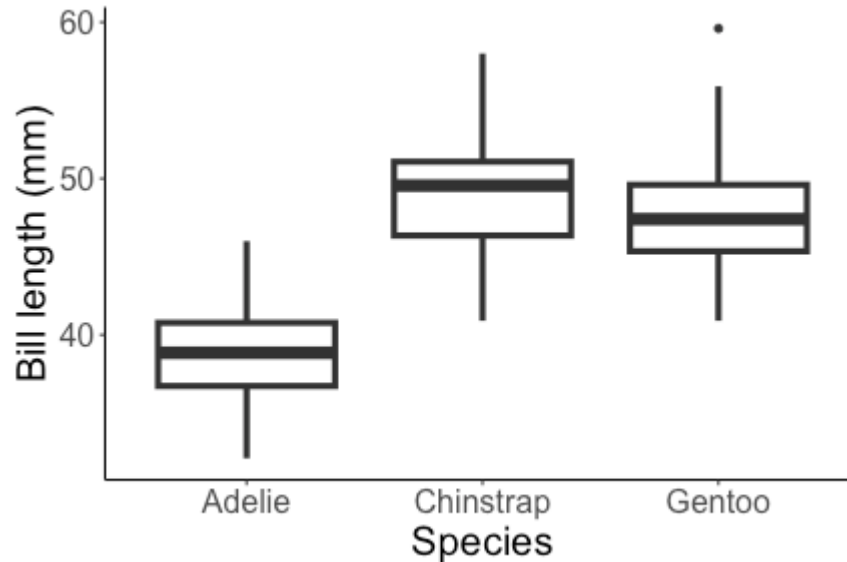


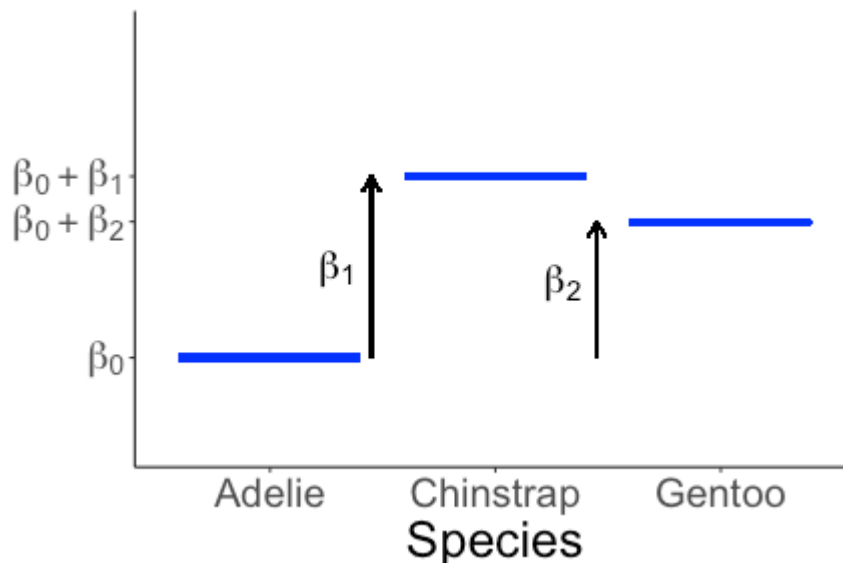
# Regression and hypothesis testing with a categorical explanatory variable

## Categorical predictor



- + How do we model the relationship between species and bill length?
- + Can we test for a relationship between bill length and species?

# Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

How can I write this more concisely?

## Indicator variables

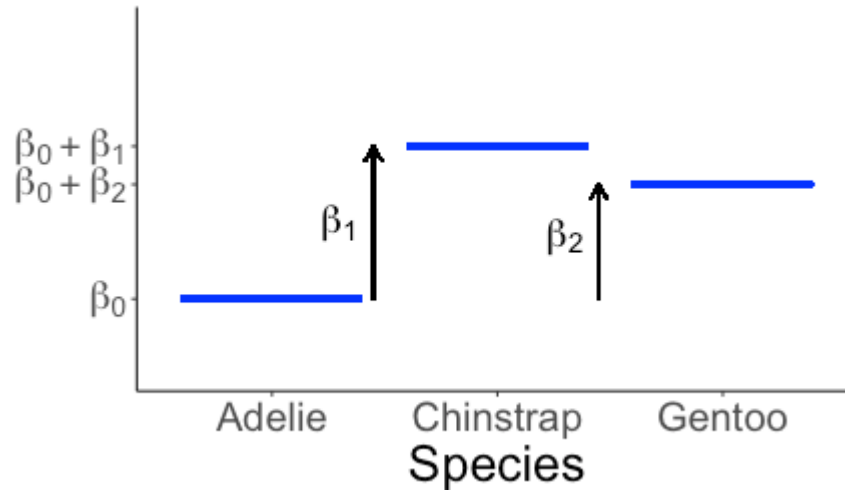
$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Then:

| Species   | IsChinstrap | IsGentoo |
|-----------|-------------|----------|
| Adelie    | 0           | 0        |
| Chinstrap | 1           | 0        |
| Gentoo    | 0           | 1        |

# Indicator variables



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

## Model assumptions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

What model assumptions do you think we make?

## Model assumptions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

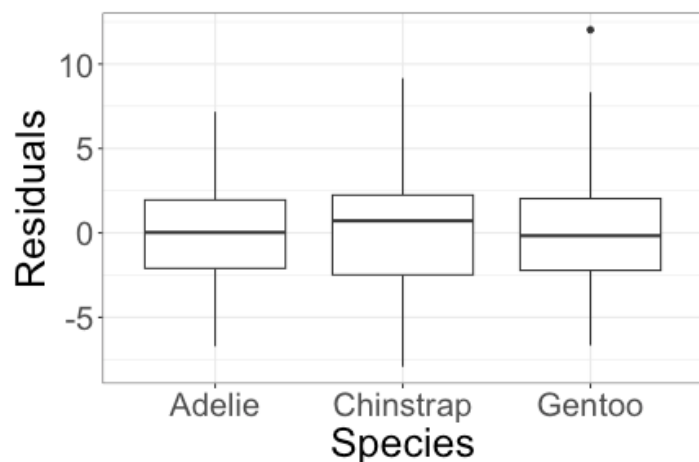
where  $\varepsilon \sim N(0, \sigma_\varepsilon^2)$  and the  $\varepsilon$  are independent of each other.

- + No notion of shape assumption with single categorical predictor
- + Still assume constant variance, normality, independence, and randomness

How should we assess each of these assumptions?

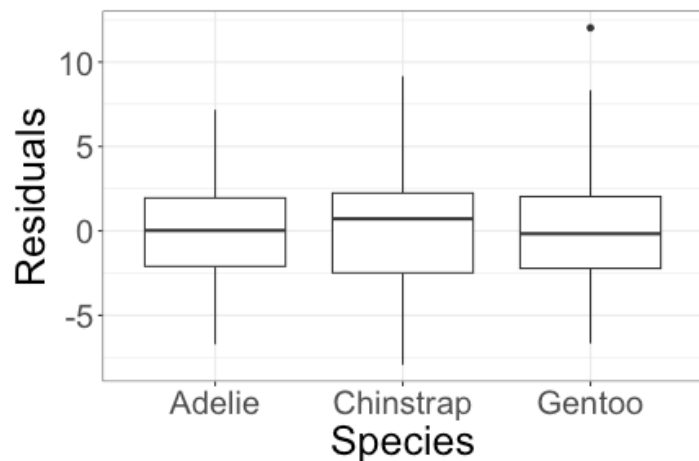
## Assessing constant variance

For a single categorical predictor, we can create a boxplot of residuals for each group, and calculate the standard deviation of residuals in each group.



| species   | residual_std_dev |
|-----------|------------------|
| Adelie    | 2.66             |
| Chinstrap | 3.34             |
| Gentoo    | 3.11             |

## Assessing constant variance



| species   | residual_std_dev |
|-----------|------------------|
| Adelie    | 2.66             |
| Chinstrap | 3.34             |
| Gentoo    | 3.11             |

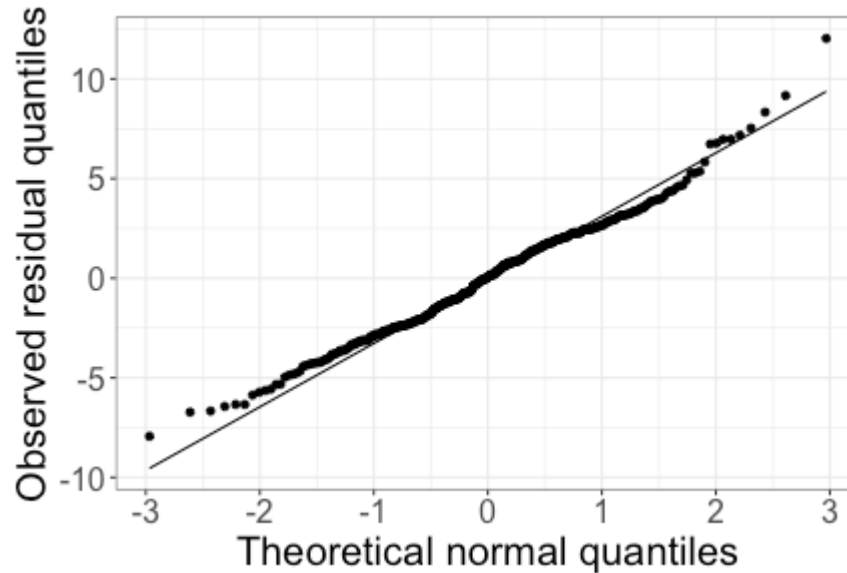
Looking for similar variability between groups. Rule of thumb: not too concerned about constant variance if

$$\frac{\text{largest standard deviation}}{\text{smallest standard deviation}} < 2$$

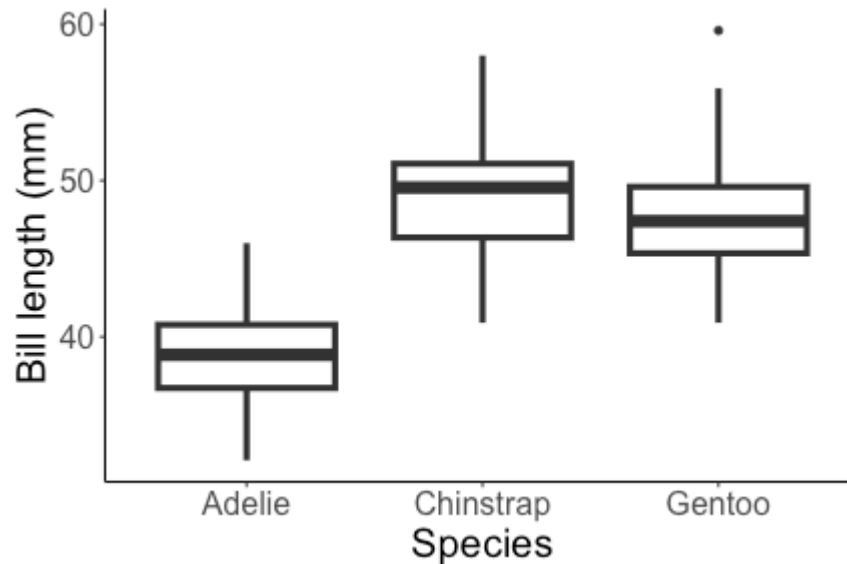
Here:  $\frac{3.34}{2.66} = 1.26 < 2$

# Assessing normality

Can use a QQ plot again:



# Statistical inference



Does it look like there is a relationship between species and bill length?

# Hypothesis testing

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

Can I translate this into null and alternative hypotheses about one or more model parameters?

# Hypothesis testing

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

Can I translate this into null and alternative hypotheses about one or more model parameters?

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

## Full and reduced models

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$$

Reduced model ( $H_0$ ):

← model under  $H_0$  (i.e.  $\beta_1 = \beta_2 = 0$ )  
(no relationship w/ species)

$$\text{bill length} = \beta_0 + \varepsilon$$

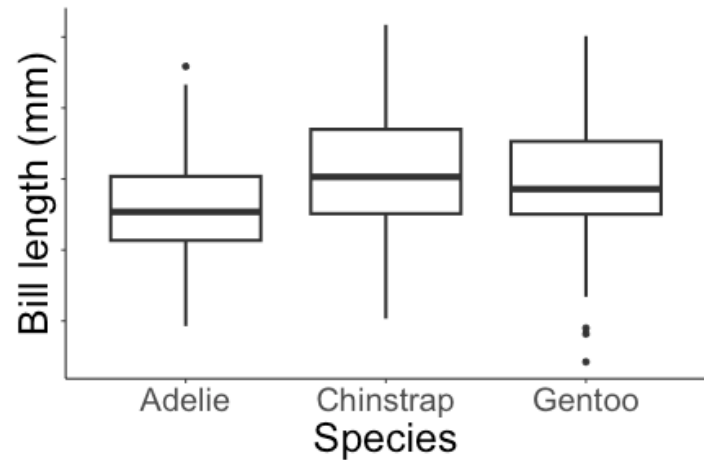
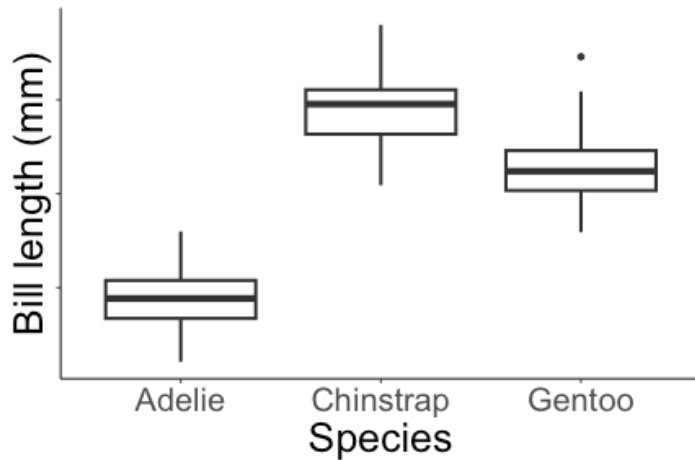
Full model ( $H_A$ ): ← model under  $H_A$

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

What do I need to test these hypotheses?

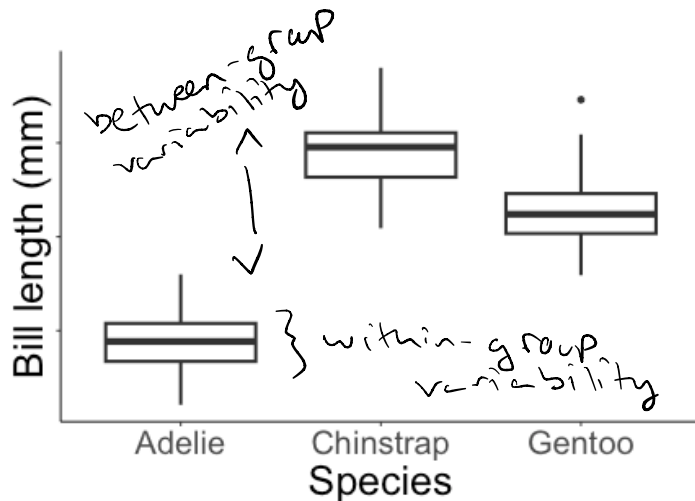
# Intuition

Which of these two plots would show more evidence of a relationship between species and bill length?

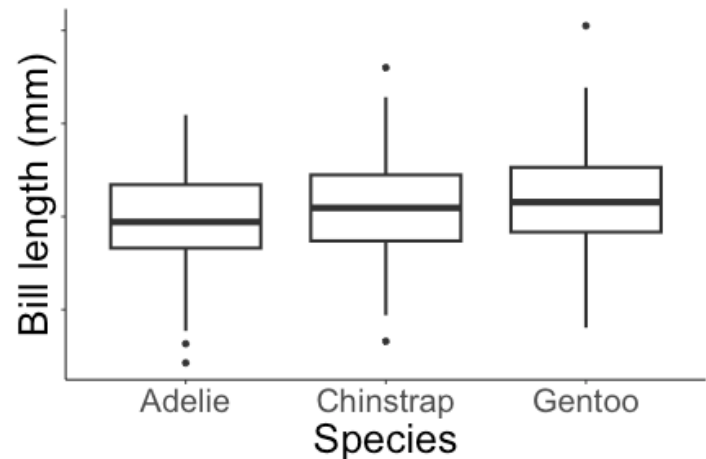


# Intuition

More evidence for a relationship when *between-group* variability is larger than *within-group* variability.



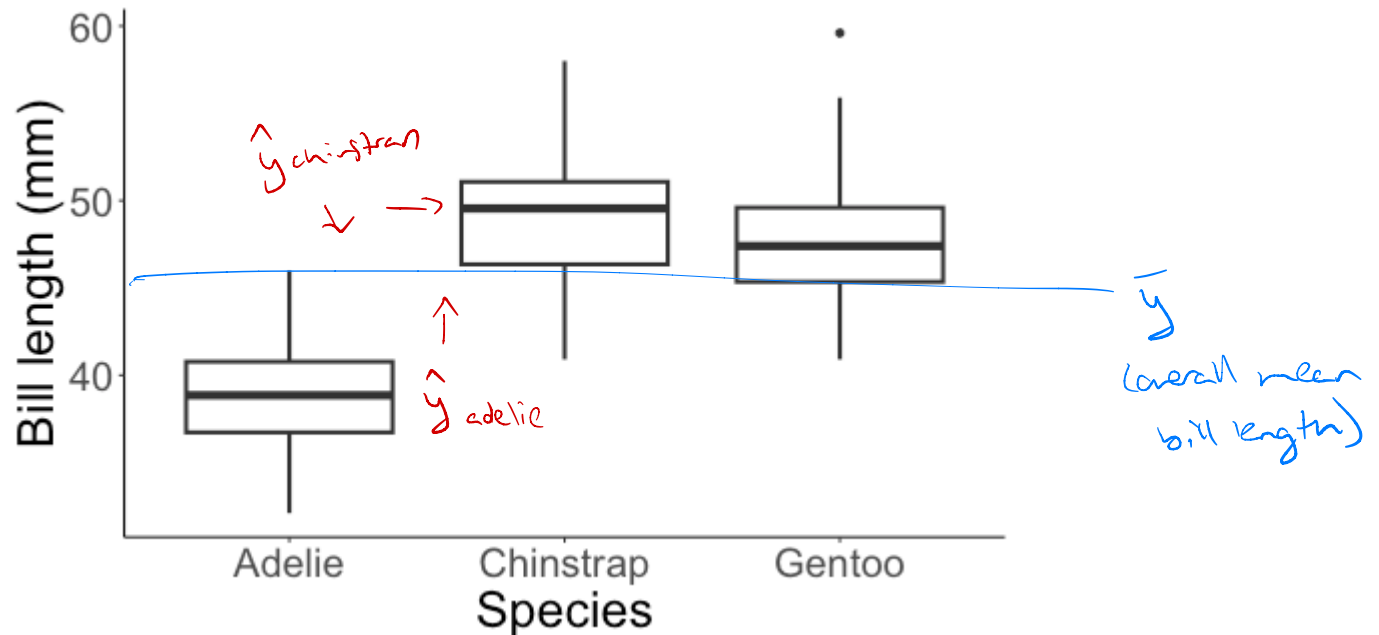
Large between-group variability, relative to within-group variability.



Small between-group variability, relative to within-group variability.

Test statistic : 
$$\frac{\text{between-group variability}}{\text{within-group variability}}$$

# Between-group vs. within-group variability



Between-group variability:  $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

Within-group variability:  $\sum_{i=1}^n (y_i - \hat{y}_i)^2$

$SS_E$

# Partitioning variability and degrees of freedom

Variability: overall variability in response

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSTotal = SSModel + SSE$$

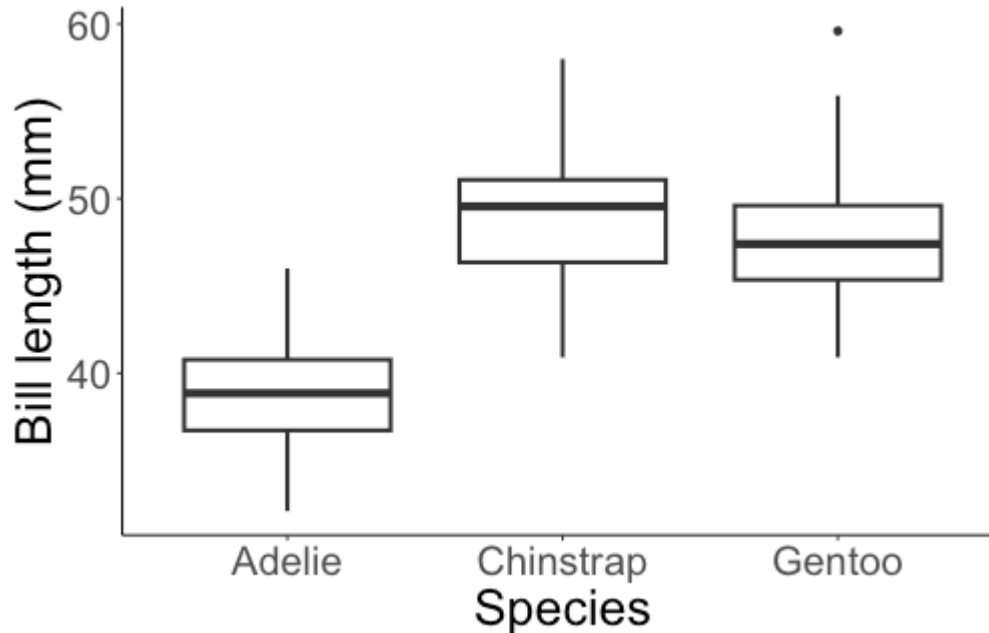
Degrees of freedom: (variability between groups) (variability w/in groups)

$$n - 1 = (p - 1) + (n - p)$$

$n = \#$  observations

$p = \#$  parameters in model  
(# of  $\beta$ s)

## Between-group vs. within-group variability



Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p - 1)}{SSE/(n - p)}$$

# Partitioning variability

Variability:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SSTotal = SSModel + SSE$$

Test statistic:

$$F = \frac{\frac{1}{p-1} \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1} SSModel}{\frac{1}{n-p} SSE} = \frac{MSModel}{MSE}$$

# Analysis of variance

Test statistic:

$$F = \frac{\frac{1}{p-1} \sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1} SS_{Model}}{\frac{1}{n-p} SSE} = \frac{MS_{Model}}{MSE}$$

Analysis of variance (ANOVA) table:

| Source   | df      | SS            | MS                                    | F                            |
|----------|---------|---------------|---------------------------------------|------------------------------|
| Model    | $p - 1$ | $SS_{Model}$  | $MS_{Model} = \frac{SS_{Model}}{p-1}$ | $F = \frac{MS_{Model}}{MSE}$ |
| Residual | $n - p$ | $SSE$         | $MSE = \frac{SSE}{n-p}$               |                              |
| Total    | $n - 1$ | $SST_{total}$ |                                       |                              |

333 penguins

$$p = 3 \quad (\beta_0, \beta_1, \beta_2)$$

## Example

$$\sum_{i=1}^n (\widehat{\text{bill length}}_i - \overline{\text{bill length}})^2 = 7015.4 \quad \leftarrow SS_{\text{Model}}$$

$$\sum_{i=1}^n (\text{bill length}_i - \widehat{\text{bill length}}_i)^2 = 2913.5 \quad \leftarrow SS_E$$

7015.4 / 2

| Source   | df  | SS     | MS     | F      |
|----------|-----|--------|--------|--------|
| Model    | 2   | 7015.4 | 3507.7 | 397.25 |
| Residual | 330 | 2913.5 | 8.83   |        |
| Total    | 332 | 9928.9 |        |        |

(n-1)

$$7015.4 + 2913.5$$

$$\frac{2913.5}{330}$$

$$\frac{3507.7}{8.83}$$

## Class activity

Work on the class activity (handout).